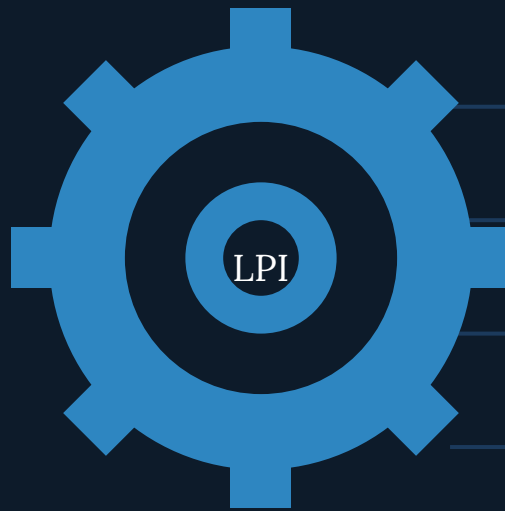




LPIC-1 (101 + 102)

Interview Questions & Answers

Complete Linux Administrator Preparation Guide

300

Questions

10

Topics

Beginner to

Advanced

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LPIC-1 Certification Preparation | 2025 Edition

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PREFACE & INTRODUCTION

Welcome to the LPIC-1 (101 + 102) Interview Questions & Answers guide — your comprehensive companion for Linux Professional Institute Certification preparation.

The Linux Professional Institute Certification Level 1 (LPIC-1) is the first certification in LPI's multi-level Linux professional certification program. The LPIC-1 validates the candidate's ability to perform maintenance tasks on the command line, install and configure a computer running Linux, and configure basic networking.

This guide contains 300 carefully crafted interview questions and answers spanning all major LPIC-1 topics, organized from beginner to advanced level. Whether you are preparing for the actual LPI examination or a technical interview for a Linux System Administrator role, this book will help you build strong foundational knowledge and advanced practical understanding.

Topics Covered:

- Linux Basics & GNU/Unix Commands • Devices, Filesystems & FHS
- Shells, Scripting & Desktop Environments • Administrative Tasks
- Essential System Services • Networking Fundamentals • Security

How to use this guide: Read through questions progressively from basic to advanced. Try to answer each question yourself before reading the provided answer. Use the questions as flashcards for revision. Focus on understanding concepts rather than memorizing answers.

Best of luck in your LPIC-1 journey!

— *Akhilesh Chaudhari*

Section 1: Linux Basics

Question 1

What is Linux?

Answer:

Linux is an open-source, Unix-like operating system kernel created by Linus Torvalds in 1991. It is freely available, highly stable, and used in servers, cloud computing, embedded systems, and desktops because of its security, flexibility, and strong community support.

Question 2

What is the Linux kernel?

Answer:

The Linux kernel is the core component of the Linux operating system. It manages system resources such as CPU, memory, and input/output devices. It acts as a bridge between hardware and software applications.

Question 3

What is an open-source operating system?

Answer:

An open-source operating system is one whose source code is freely available for anyone to view, modify, and distribute. Linux is a prime example — its source code is available under the GNU General Public License (GPL).

Question 4

Who created Linux and when?

Answer:

Linux was created by Linus Torvalds, a Finnish-American software engineer, in 1991. He initially developed it as a personal project and later released it to the public, where it grew with contributions from thousands of developers worldwide.

Question 5

What is the difference between Linux and Unix?

Answer:

Unix is a proprietary operating system developed in the 1970s at AT&T; Bell Labs. Linux is inspired by Unix but was written from scratch and is open-source. Both share similar concepts and commands, but Linux is free while most Unix systems are commercial.

Question 6

What is a Linux distribution (distro)?

Answer:

A Linux distribution is a packaged version of Linux that includes the kernel, system libraries, utilities, and software. Examples include Ubuntu, CentOS, Debian, Fedora, and openSUSE. Each distro targets different use cases such as servers, desktops, or embedded systems.

Question 7

What is the GNU Project?

Answer:

The GNU Project was initiated by Richard Stallman in 1983 with the goal of creating a free Unix-like operating system. GNU provides many tools and utilities (like gcc, bash, glibc) used alongside the Linux kernel to form a complete operating system often called GNU/Linux.

Question 8

What is POSIX?

Answer:

POSIX (Portable Operating System Interface) is a family of standards specified by IEEE that defines the API for software compatibility with Unix-like systems. Linux follows POSIX standards, ensuring portability of programs across compatible systems.

Question 9

What is a shell in Linux?

Answer:

A shell is a command-line interpreter that allows users to interact with the operating system by typing commands. It reads user input, interprets it, and executes corresponding programs. Examples include Bash, sh, zsh, and ksh.

Question 10

What is Bash?

Answer:

Bash (Bourne Again Shell) is the most widely used shell in Linux. It is an enhanced version of the original Bourne shell (sh) and supports scripting, command history, tab completion, and many built-in commands. It is the default shell on most Linux distributions.

Question 11

What is a terminal in Linux?

Answer:

A terminal (also called a terminal emulator) is a program that provides a text-based interface to the shell. It allows users to type commands and see output. Examples include GNOME Terminal, xterm, and Konsole.

Question 12

What is the difference between CLI and GUI?

Answer:

CLI (Command Line Interface) allows users to interact with the system by typing text commands, offering more control and efficiency. GUI (Graphical User Interface) uses windows, icons, and menus for interaction, making it more user-friendly for beginners.

Question 13

What is the root user in Linux?

Answer:

The root user is the superuser or administrator account in Linux with unrestricted access to all system resources and commands. It can modify system files, install software, and manage users. The root account should be used cautiously to avoid accidental damage.

Question 14

What is sudo in Linux?

Answer:

sudo (superuser do) allows a permitted user to execute a command as the superuser or another user, as defined in the `/etc/sudoers` file. It provides controlled privilege escalation without requiring the root password to be shared.

Question 15

What is the home directory in Linux?

Answer:

The home directory is a personal directory assigned to each user, typically located at `/home/username`. It stores personal files, configuration files, and user-specific settings. The root user's home directory is `/root`.

Question 16

What is the `/etc` directory used for?

Answer:

`/etc` (et cetera) contains system-wide configuration files and initialization scripts. Important files like `/etc/passwd` (user accounts), `/etc/fstab` (filesystems), and `/etc/hostname` are stored here. Most system configuration is managed through files in this directory.

Question 17

What does the `pwd` command do?

Answer:

`pwd` (print working directory) displays the full path of the current directory you are in. For example, running `pwd` while in the home directory of user 'john' would output `/home/john`.

Question 18

What is a process in Linux?

Answer:

A process is a running instance of a program. Each process has a unique Process ID (PID), and it uses system resources like CPU and memory. Linux supports multitasking, allowing multiple processes to run simultaneously.

Question 19

What is a daemon in Linux?

Answer:

A daemon is a background process that runs continuously without user interaction. Daemons typically start at boot time and provide system services. Examples include sshd (SSH service), httpd (web server), and crond (cron scheduler).

Question 20

What is the init system in Linux?

Answer:

The init system is the first process started by the kernel (PID 1) and is responsible for initializing the system and managing services. Modern Linux distributions use systemd as the init system, replacing the older SysVinit.

Question 21

What is systemd?

Answer:

systemd is a modern init system and service manager for Linux. It starts and manages system services, handles dependencies between services, and provides logging via journald. Commands like systemctl are used to manage services with systemd.

Question 22

What is a runlevel in Linux?

Answer:

A runlevel defines the state of the machine in terms of which services are running. Traditional SysV init defines runlevels 0–6 (0=halt, 1=single-user, 3=multi-user text, 5=multi-user GUI, 6=reboot). systemd uses targets instead of runlevels.

Question 23

What is the PATH variable?

Answer:

PATH is an environment variable that specifies a list of directories the shell searches to find executable programs. When you type a command, the shell looks through each directory in PATH to locate the binary. Example: /usr/local/bin:/usr/bin:/bin.

Question 24

How do you view environment variables?

Answer:

You can view all environment variables using the 'env' or 'printenv' commands. To view a specific variable, use 'echo \$VARIABLE_NAME'. For example, 'echo \$PATH' displays the current PATH variable.

Question 25

What is a package manager in Linux?

Answer:

A package manager is a tool that automates installing, updating, and removing software packages. Different distributions use different managers: APT for Debian/Ubuntu, YUM/DNF for Red Hat/CentOS/Fedora, and zypper for openSUSE.

Question 26

What is the difference between a process and a thread?

Answer:

A process is an independent program with its own memory space and resources. A thread is a smaller unit of execution within a process that shares the same memory space. Threads are lighter and faster to create than processes.

Question 27

What is virtual memory in Linux?

Answer:

Virtual memory is a memory management technique that creates an abstraction of the physical memory. It allows programs to use more memory than physically available by using disk space (swap) as an extension of RAM.

Question 28

What is swap space in Linux?

Answer:

Swap space is a portion of the hard disk used as virtual memory when physical RAM is full. The kernel moves inactive memory pages to swap to free up RAM for active processes. It can be a dedicated swap partition or a swap file.

Question 29

What is the /proc filesystem?

Answer:

/proc is a virtual filesystem that provides information about running processes and kernel state. It does not contain real files but shows runtime system information. For example, /proc/cpuinfo contains CPU details and /proc/meminfo shows memory usage.

Question 30

What is the difference between hard links and symbolic links?

Answer:

A hard link is a direct reference to the same inode (data) on disk. Deleting the original file does not affect hard links. A symbolic (soft) link is a pointer to another file by path name. If the original file is deleted, the symlink becomes broken.

Section 2: GNU & Unix Commands

Question 31

What does the ls command do?

Answer:

ls lists the contents of a directory. Common options: -l (long format with permissions), -a (show hidden files), -h (human-readable sizes), -r (reverse order), -t (sort by modification time). Example: 'ls -lah /home' shows detailed listing of /home.

Question 32

How do you copy files in Linux?

Answer:

The cp command copies files and directories. Syntax: 'cp source destination'. Options: -r or -R (recursive, for directories), -p (preserve permissions and timestamps), -i (interactive, ask before overwrite). Example: 'cp -r /source /destination'.

Question 33

How do you move or rename files in Linux?

Answer:

The mv command moves or renames files and directories. 'mv oldname newname' renames a file. 'mv file /new/path/' moves it. Unlike cp, mv does not duplicate the file — it simply changes its location or name.

Question 34

How do you delete files and directories?

Answer:

Use rm to delete files: 'rm filename'. For directories, use 'rm -r dirname' (recursive). Use -f to force deletion without prompting. Caution: 'rm -rf /' would delete the entire filesystem — always double-check before using -rf.

Question 35

What does the cat command do?

Answer:

cat (concatenate) displays file contents, concatenates multiple files, or creates new files. 'cat file.txt' displays content. 'cat file1 file2 > merged.txt' merges files. 'cat > newfile.txt' creates a new file (Ctrl+D to save).

Question 36

What is the grep command?

Answer:

grep searches for patterns in files or input. 'grep pattern file' prints lines matching the pattern. Common options: -i (case-insensitive), -r (recursive), -n (show line numbers), -v (invert match). Example: 'grep -in error /var/log/syslog'.

Question 37

What does the find command do?

Answer:

find searches for files and directories in a directory hierarchy. Example: 'find /home -name '*.txt'' finds all .txt files under /home. Options include -type (f=file, d=directory), -mtime (modification time), -size, and -exec to run commands on results.

Question 38

What is the chmod command?

Answer:

chmod (change mode) changes file or directory permissions. You can use symbolic mode ('chmod u+x file' adds execute for owner) or octal mode ('chmod 755 file' sets rwxr-xr-x). Permissions: r=4, w=2, x=1 for owner (u), group (g), others (o).

Question 39

What is the chown command?

Answer:

chown (change owner) changes the ownership of a file or directory. Syntax: 'chown owner:group file'. Example: 'chown alice:developers project/' changes owner to alice and group to developers. Use -R for recursive changes.

Question 40

What does the ps command do?

Answer:

ps (process status) displays information about active processes. 'ps aux' shows all processes with CPU and memory usage. 'ps -ef' shows a full-format listing. Often used with grep: 'ps aux | grep nginx' to find a specific process.

Question 41

What is the top command?

Answer:

top is an interactive real-time system monitor showing CPU usage, memory usage, and running processes. Press 'q' to quit, 'k' to kill a process, 'M' to sort by memory, and 'P' to sort by CPU. 'htop' is an enhanced, more user-friendly version.

Question 42

What does the kill command do?

Answer:

kill sends a signal to a process, usually to terminate it. 'kill PID' sends SIGTERM (graceful termination). 'kill -9 PID' sends SIGKILL (force kill). Use 'killall processname' to kill all processes with a given name.

Question 43

What is the df command?

Answer:

df (disk free) reports filesystem disk space usage. 'df -h' shows human-readable sizes (KB, MB, GB). It shows total size, used space, available space, and mount point for each filesystem. Useful for monitoring disk capacity.

Question 44

What is the du command?

Answer:

du (disk usage) estimates the space used by files and directories. 'du -sh /path' shows human-readable summary of total space used. 'du -ah' shows sizes for all files. Useful for finding large directories consuming disk space.

Question 45

What does the tar command do?

Answer:

tar (tape archive) creates and extracts archive files. 'tar -czf archive.tar.gz /path' creates a gzip-compressed archive. 'tar -xzf archive.tar.gz' extracts it. Options: -c (create), -x (extract), -z (gzip), -j (bzip2), -v (verbose), -f (file).

Question 46

What is the pipe operator (|) in Linux?

Answer:

The pipe operator sends the output of one command as input to another. Example: 'ps aux | grep nginx' passes the output of ps to grep. Pipes allow chaining commands to build powerful one-liners without creating temporary files.

Question 47

What is input/output redirection?

Answer:

Redirection changes where command input comes from or where output goes. '>' redirects stdout to a file (overwrite). '>>' appends to a file. '<' reads input from a file. '2>' redirects stderr. '2>&1' merges stderr into stdout.

Question 48

What does the head command do?

Answer:

head displays the first lines of a file. By default, it shows the first 10 lines. 'head -n 20 file.txt' shows the first 20 lines. Useful for quickly previewing large log files or text documents.

Question 49

What does the tail command do?

Answer:

tail displays the last lines of a file. 'tail -n 20 file.txt' shows the last 20 lines. 'tail -f file.txt' follows the file in real time, making it ideal for monitoring log files as new entries are added.

Question 50

What is the sed command?

Answer:

sed (stream editor) is used to filter and transform text. It can search, replace, delete, and insert text. Example: 'sed 's/old/new/g' file.txt' replaces all occurrences of 'old' with 'new'. The -i flag edits files in place.

Question 51

What is the awk command?

Answer:

awk is a powerful text-processing tool used to search, filter, and transform structured text. It processes text line by line and splits each line into fields. Example: 'awk '{print \$1, \$3}' file.txt' prints the 1st and 3rd columns.

Question 52

What does the wc command do?

Answer:

wc (word count) counts lines, words, and characters in a file. 'wc -l file' counts lines, 'wc -w file' counts words, and 'wc -c file' counts bytes. Example: 'cat /etc/passwd | wc -l' counts the number of user accounts.

Question 53

What is the sort command?

Answer:

sort arranges lines of text alphabetically or numerically. 'sort file.txt' sorts alphabetically. Options: -n (numeric sort), -r (reverse), -k (sort by column), -u (remove duplicates). Example: 'sort -n -k2 data.txt' sorts by 2nd column numerically.

Question 54

What does the uniq command do?

Answer:

uniq filters out adjacent duplicate lines from sorted input. 'uniq file.txt' removes consecutive duplicates. Use -c to count occurrences, -d to show only duplicates. It is often combined with sort: 'sort file | uniq -c | sort -rn'.

Question 55

What is the echo command?

Answer:

echo displays a line of text or the value of variables. 'echo Hello World' prints 'Hello World'. 'echo \$HOME' prints the home directory path. The -n flag suppresses the trailing newline, and -e enables interpretation of escape sequences.

Question 56

What is the diff command?

Answer:

diff compares files line by line and shows differences. 'diff file1 file2' shows what needs to change in file1 to make it match file2. Use -u for unified format (used in patches). 'diff -r dir1 dir2' compares directories recursively.

Question 57

What does the which command do?

Answer:

which shows the full path of a command's executable. 'which python3' might return '/usr/bin/python3'. This is useful for determining which version of a program will run when you type its name, especially if multiple versions exist.

Question 58

What is the whereis command?

Answer:

whereis locates the binary, source, and man page files for a command. 'whereis ls' returns locations of the ls binary, its source code (if available), and its manual page. It searches standard directories defined for each type of file.

Question 59

What does the man command do?

Answer:

man (manual) displays the manual page for a command. 'man ls' opens the documentation for ls. Manual pages are organized into sections (1=user commands, 5=file formats, 8=sysadmin commands). Use 'man -k keyword' to search by keyword.

Question 60

What is the alias command?

Answer:

alias creates a shortcut for a command. 'alias ll='ls -lah' creates an alias 'll' that runs 'ls -lah'. Aliases defined in the terminal session are temporary; to make them permanent, add them to ~/.bashrc or ~/.bash_profile.

Question 61

What is the history command?

Answer:

history displays a list of previously executed commands. Each command has a number; you can re-run it with '!number'. 'history | grep ssh' filters history for ssh commands. The history is stored in ~/.bash_history by default.

Question 62

What does the cut command do?

Answer:

cut removes sections from lines of text. It can cut by character position, byte, or delimiter. 'cut -d: -f1 /etc/passwd' extracts the first field (username) from /etc/passwd, using colon as the delimiter.

Question 63

What does the tr command do?

Answer:

tr (translate or delete) translates or deletes characters. 'echo hello | tr 'a-z' 'A-Z' converts lowercase to uppercase. 'tr -d '\n' removes newlines. It works only with standard input, not files directly.

Question 64

What is the xargs command?

Answer:

xargs builds and executes command lines from standard input. It reads items from stdin and passes them as arguments to a command. Example: 'find . -name '*.log' | xargs rm' deletes all .log files found by find.

Question 65

What is the tee command?

Answer:

tee reads from stdin and writes to both stdout and a file simultaneously. 'command | tee output.txt' displays command output on screen and also saves it to output.txt. Use -a to append to an existing file instead of overwriting.

Question 66

What does the ln command do?

Answer:

ln creates links between files. 'ln file hardlink' creates a hard link. 'ln -s target linkname' creates a symbolic (soft) link. Symbolic links are like shortcuts, while hard links are direct references to the same disk data.

Question 67

What is the touch command?

Answer:

touch creates an empty file if it doesn't exist, or updates the access and modification timestamps of an existing file. 'touch newfile.txt' creates newfile.txt. It is commonly used in scripts to create placeholder files.

Question 68

What does the file command do?

Answer:

file determines the type of a file by examining its content, not just its extension. 'file script.sh' might return 'Bourne-Again shell script, ASCII text executable'. This is useful when file extensions are missing or misleading.

Question 69

What is the stat command?

Answer:

stat displays detailed information about a file or filesystem, including size, inode number, permissions, owner, group, and timestamps (access, modify, change). 'stat file.txt' shows all metadata for file.txt.

Question 70

What does the uname command do?

Answer:

uname prints system information. 'uname -a' shows all information including kernel name, hostname, kernel version, machine hardware, and operating system. 'uname -r' shows just the kernel version, which is useful for checking compatibility.

Section 3: Devices, Linux Filesystems & Filesystem Hierarchy

Question 71

What is the Linux Filesystem Hierarchy Standard (FHS)?

Answer:

The FHS defines the directory structure and directory contents in Linux systems. It standardizes where files and directories should be placed, ensuring consistency across distributions. Key directories include `/`, `/bin`, `/etc`, `/home`, `/var`, `/usr`, `/tmp`, and `/dev`.

Question 72

What is the purpose of the `/bin` directory?

Answer:

`/bin` (binary) contains essential command binaries needed for the system to function in single-user mode. Commands like `ls`, `cp`, `mv`, `rm`, `bash`, and `cat` are stored here. All users can execute files in `/bin`.

Question 73

What is the `/sbin` directory?

Answer:

`/sbin` (system binary) contains essential system administration binaries used by the root user. Commands like `fdisk`, `ifconfig`, `init`, and `mkfs` are stored here. Regular users can view but typically cannot execute these commands.

Question 74

What is the `/usr` directory?

Answer:

`/usr` (Unix System Resources) contains user-land applications, libraries, and documentation. It includes `/usr/bin` (non-essential commands), `/usr/sbin` (non-essential system admin commands), `/usr/lib` (libraries), and `/usr/share` (shared data).

Question 75

What is the `/var` directory used for?

Answer:

`/var` (variable) stores files that change frequently during system operation. This includes logs (`/var/log`), mail spools (`/var/mail`), print spools (`/var/spool`), and lock files (`/var/lock`). System activity data is typically stored here.

Question 76

What is the `/tmp` directory?

Answer:

`/tmp` stores temporary files created by programs during runtime. Files here may be automatically deleted at boot or after a set period. Programs should not rely on data persisting across reboots in `/tmp`.

Question 77

What is the `/dev` directory?

Answer:

`/dev` (devices) contains device files that represent hardware and virtual devices. Block devices (disks) appear as `/dev/sda`, `/dev/nvme0n1`. Character devices (terminals, serial ports) appear as `/dev/tty`. Special files `/dev/null`, `/dev/zero`, and `/dev/random` are also here.

Question 78

What is the `/proc` directory?

Answer:

`/proc` is a virtual filesystem mounted at boot that provides an interface to kernel data structures. It contains information about running processes (`/proc/PID/`), system memory (`/proc/meminfo`), CPU (`/proc/cpuinfo`), and kernel parameters (`/proc/sys/`).

Question 79

What is the `/sys` directory?

Answer:

`/sys` is a virtual filesystem (sysfs) that exposes information about devices and drivers to user space. It provides a structured view of the kernel's device model and allows user-space programs to query and configure hardware.

Question 80

What is the `/boot` directory?

Answer:

`/boot` contains files required to boot the system, including the Linux kernel (`vmlinuz`), initial RAM disk (`initrd/initramfs`), and the GRUB bootloader files. Keeping `/boot` on a separate partition is common for encrypted systems.

Question 81

What is the `/mnt` directory?

Answer:

`/mnt` is a generic mount point for temporarily mounting filesystems. Administrators use it to manually mount external drives, network shares, or ISO images. For example: `'mount /dev/sdb1 /mnt'`.

Question 82

What is the `/media` directory?

Answer:

`/media` is used for automatically mounted removable media such as USB drives, CD-ROMs, and external hard drives. Modern desktop environments automatically create subdirectories here when removable devices are connected.

Question 83

What is the `/opt` directory?

Answer:

`/opt` (optional) is used for installing optional or third-party software that is not part of the standard distribution. Commercial applications and manually installed software typically go here. Each package usually has its own subdirectory.

Question 84

What are inodes in Linux?

Answer:

An inode is a data structure that stores metadata about a file, including permissions, ownership, timestamps, size, and pointers to data blocks on disk. Every file and directory has an inode. The filename is stored in the directory, not the inode.

Question 85

What is a block device?

Answer:

A block device is a hardware device that stores and retrieves data in fixed-size blocks. Examples include hard drives (`/dev/sda`), SSDs, and optical drives. Block devices support random access and are typically formatted with a filesystem.

Question 86

What is a character device?

Answer:

A character device handles data as a stream of characters (byte by byte) without buffering. Examples include terminals (`/dev/tty`), serial ports (`/dev/ttyS0`), and `/dev/null`. They do not support random access like block devices.

Question 87

What is `/dev/null`?

Answer:

`/dev/null` is a special file that discards all data written to it and returns end-of-file when read. It is used to suppress output: `'command > /dev/null 2>&1'` discards both stdout and stderr. Sometimes called 'the bit bucket'.

Question 88

What is a filesystem in Linux?

Answer:

A filesystem is a method of organizing and storing files on a storage device. It manages how data is stored, retrieved, and organized. Linux supports many filesystems including ext4, xfs, btrfs, ntfs, fat32, and more.

Question 89

What is ext4?

Answer:

ext4 (fourth extended filesystem) is the default filesystem for many Linux distributions. It supports large files and volumes, journaling (which prevents data corruption after crashes), extended attributes, and backward compatibility with ext2/ext3.

Question 90

What is XFS?

Answer:

XFS is a high-performance journaling filesystem developed by SGI. It is optimized for large files and high I/O workloads. It is the default filesystem in Red Hat Enterprise Linux (RHEL) 7+ and is well-suited for database and file servers.

Question 91

What is Btrfs?

Answer:

Btrfs (B-tree filesystem) is a modern copy-on-write filesystem for Linux. It supports snapshots, data integrity checking, RAID, online resizing, and subvolumes. It is designed to replace ext4 and is used by default in some distributions like openSUSE.

Question 92

What is journaling in a filesystem?

Answer:

Journaling is a technique where filesystem changes are written to a journal (log) before being committed to the main filesystem. If the system crashes, the journal can be replayed to restore consistency, preventing data corruption.

Question 93

What is the mount command?

Answer:

mount attaches a filesystem to the directory tree at a specified mount point. 'mount /dev/sdb1 /mnt/data' mounts the partition. 'mount -t ext4' specifies the filesystem type. Running 'mount' without arguments shows all currently mounted filesystems.

Question 94

What is the umount command?

Answer:

umount detaches a mounted filesystem. 'umount /mnt/data' unmounts the device at /mnt/data. You cannot unmount a filesystem that is currently in use (open files or processes). Use 'lsof /mnt/data' to find blocking processes.

Question 95

What is `/etc/fstab`?

Answer:

`/etc/fstab` (filesystem table) is a configuration file that defines how disk partitions and other filesystems should be mounted at boot. Each line specifies: device, mount point, filesystem type, mount options, dump flag, and fsck order.

Question 96

What is `fdisk`?

Answer:

`fdisk` is a command-line tool for viewing and manipulating disk partition tables on MBR disks. '`fdisk -l`' lists all disk partitions. '`fdisk /dev/sda`' opens an interactive menu to create, delete, or modify partitions.

Question 97

What is `parted`?

Answer:

`parted` is a disk partitioning tool that supports both MBR and GPT partition tables, making it more versatile than `fdisk`. It supports resizing partitions and is scriptable. '`parted /dev/sda print`' displays partition information.

Question 98

What is `gdisk`?

Answer:

`gdisk` (GPT `fdisk`) is a partition editor for GPT (GUID Partition Table) disks. It is used on systems with UEFI firmware. Like `fdisk`, it provides an interactive menu for creating and managing partitions on modern disks.

Question 99

What is `mkfs`?

Answer:

`mkfs` (make filesystem) creates a filesystem on a device or partition. '`mkfs.ext4 /dev/sdb1`' formats the partition with ext4. '`mkfs.xfs /dev/sdc1`' formats with XFS. Always ensure you have selected the correct device before formatting.

Question 100

What is `fsck`?

Answer:

`fsck` (filesystem check) is used to check and repair filesystem inconsistencies. '`fsck /dev/sda1`' checks the partition. It should only be run on unmounted filesystems. On most systems, `fsck` runs automatically at boot if the filesystem is marked as dirty.

Question 101

What is LVM (Logical Volume Management)?

Answer:

LVM is a storage management framework that provides an abstraction layer over physical disks. It allows creating logical volumes that can be resized, snapshots that can be taken, and storage from multiple disks combined. Components: Physical Volumes (PV), Volume Groups (VG), Logical Volumes (LV).

Question 102

What is RAID in Linux?

Answer:

RAID (Redundant Array of Independent Disks) combines multiple physical disks into one logical unit for performance or redundancy. Software RAID in Linux is managed using mdadm. Common levels: RAID 0 (striping), RAID 1 (mirroring), RAID 5 (distributed parity).

Question 103

What is swap and how is it created?

Answer:

Swap is disk space used as virtual memory. To create a swap partition: use fdisk to create it, run 'mkswap /dev/sdX', then 'swapon /dev/sdX'. To create a swap file: 'dd if=/dev/zero of=/swapfile bs=1G count=2', 'mkswap /swapfile', 'swapon /swapfile'.

Question 104

What does the lsblk command do?

Answer:

lsblk (list block devices) displays information about all block devices in a tree format. It shows device names, sizes, mount points, and types. 'lsblk -f' also shows filesystem type and UUID. Very useful for quickly understanding disk layout.

Question 105

What is a UUID and why is it used in Linux?

Answer:

UUID (Universally Unique Identifier) is a unique string assigned to filesystems during creation. In /etc/fstab, UUIDs are used instead of device names (like /dev/sda1) because device names can change between boots, while UUIDs remain stable.

Question 106

What is the blkid command?

Answer:

blkid displays block device attributes including UUID, filesystem type, and label. 'blkid' lists all block devices, 'blkid /dev/sda1' shows details for a specific partition. Useful for finding UUIDs to use in /etc/fstab.

Question 107

What is a mount option 'noexec'?

Answer:

The 'noexec' mount option prevents execution of binaries on the mounted filesystem. This is a security measure used on partitions like /tmp or /home to prevent users from running malicious executables placed there.

Question 108

What is the difference between MBR and GPT?

Answer:

MBR (Master Boot Record) is an older partitioning scheme limited to disks up to 2TB and a maximum of 4 primary partitions. GPT (GUID Partition Table) supports disks larger than 2TB, up to 128 partitions, and provides better data integrity with CRC checksums.

Question 109

What is /etc/mtab?

Answer:

/etc/mtab is a file that lists currently mounted filesystems, similar to /proc/mounts. It is updated by the mount and umount commands. On modern systems, it is often a symlink to /proc/self/mounts for accuracy.

Question 110

What are file permissions in Linux?

Answer:

Linux file permissions control who can read (r), write (w), and execute (x) a file. Permissions are assigned to three categories: owner (u), group (g), and others (o). They are displayed as '-rwxr-xr--' in ls -l output and represented as octal numbers like 755.

Section 4: Shells, Scripting & User Interfaces

Question 111

What is a shell script?

Answer:

A shell script is a text file containing a sequence of shell commands that are executed in order. Scripts automate repetitive tasks and are run using a shell interpreter like Bash. They begin with a shebang line (`#!/bin/bash`) to specify the interpreter.

Question 112

What is a shebang line?

Answer:

A shebang (`#!/bin/bash`) is the first line of a script that tells the kernel which interpreter to use. `#!/bin/bash` runs the script with Bash. `#!/usr/bin/env python3` runs it with Python 3. The `#!` is short for 'hash bang'.

Question 113

How do you make a script executable?

Answer:

Use `chmod` to add execute permission: `chmod +x script.sh`. Then run it with `./script.sh` from the same directory, or with the full path. Alternatively, invoke it directly: `bash script.sh` without needing execute permission.

Question 114

What are variables in Bash?

Answer:

Variables store data in Bash scripts. Assign with `'NAME=value'` (no spaces around `=`). Access with `'$NAME'` or `'${NAME}'`. Example: `'GREETING=Hello'; echo $GREETING`. Variables are untyped by default and treated as strings.

Question 115

What are special variables in Bash?

Answer:

Special variables include: `$0` (script name), `$1-$9` (positional parameters), `$#` (number of arguments), `$@` (all arguments), `$?` (exit status of last command), `$$` (PID of current shell), and `$_` (PID of last background command).

Question 116

What is an if statement in Bash?

Answer:

An if statement executes commands conditionally. Syntax: 'if [condition]; then commands; elif [condition]; then commands; else commands; fi'. Example: 'if [-f /etc/passwd]; then echo 'File exists'; fi'.

Question 117

What is a for loop in Bash?

Answer:

A for loop iterates over a list of items. 'for i in 1 2 3; do echo \$i; done' prints 1, 2, 3. 'for file in *.txt; do cat \$file; done' processes each .txt file. C-style: 'for ((i=0; i<5; i++)); do echo \$i; done'.

Question 118

What is a while loop in Bash?

Answer:

A while loop executes commands as long as a condition is true. 'while [condition]; do commands; done'. Example: 'while read line; do echo \$line; done < file.txt' reads a file line by line. Use 'break' to exit and 'continue' to skip.

Question 119

What is a case statement in Bash?

Answer:

A case statement matches a value against patterns. 'case \$var in pattern1) commands;; pattern2) commands;; *) default;; esac'. Example: 'case \$1 in start) start_service;; stop) stop_service;; *) echo 'Unknown';; esac'.

Question 120

What are functions in Bash?

Answer:

Functions group commands for reuse. Define with 'function_name() { commands; }' or 'function function_name { commands; }'. Call with 'function_name arg1 arg2'. Access args with \$1, \$2. Return values with 'return N' (0=success).

Question 121

What is command substitution in Bash?

Answer:

Command substitution captures command output and uses it as a string. Use '\$(command)' or backticks 'command'. Example: 'DATE=\$(date +%Y-%m-%d)' stores the current date. The preferred modern syntax is \$(command) over backticks.

Question 122

What is arithmetic expansion in Bash?

Answer:

Arithmetic expansion evaluates integer expressions. Use '\$((expression))'. Example: 'echo \$((5 + 3))' outputs 8. 'result=\$((a * b))' stores the product. For floating-point math, use bc or awk since Bash only handles integers.

Question 123

What is the difference between single and double quotes in Bash?

Answer:

Single quotes (') preserve the literal value of every character — no variable expansion or special characters are interpreted. Double quotes (") allow variable expansion (\$VAR) and command substitution (\$(cmd)) but prevent word splitting and glob expansion.

Question 124

What is the exit status of a command?

Answer:

The exit status (return code) is a number returned by a command when it finishes. 0 means success; non-zero means failure. Access it with \$? . Conditionals like if and while evaluate exit status. 'exit N' sets the exit status of a script.

Question 125

What is the test command in Bash?

Answer:

test evaluates conditions and returns exit status 0 (true) or 1 (false). It is equivalent to []. 'test -f file' or '[-f file]' checks if file exists and is a regular file. [[]] is an extended version with more features available in Bash.

Question 126

What are common file test operators in Bash?

Answer:

Common operators: -f (regular file), -d (directory), -e (exists), -r (readable), -w (writable), -x (executable), -s (non-empty), -L (symbolic link), -z (string is empty), -n (string is non-empty). Example: '[-d /etc] && echo 'Directory exists''.

Question 127

How do you read user input in a Bash script?

Answer:

Use the 'read' command: 'read -p 'Enter name: ' NAME'. This prompts the user and stores input in NAME. 'read -s' hides input (for passwords). 'read -t 10' times out after 10 seconds. 'read -a ARRAY' reads into an array.

Question 128

What are arrays in Bash?

Answer:

Arrays store multiple values. Define: 'FRUITS=(apple banana cherry)'. Access: '\${FRUITS[0]}' (first element), '\${FRUITS[@]}' (all elements), '\${#FRUITS[@]}' (count). Associative arrays (key-value): 'declare -A MAP; MAP[key]=value'.

Question 129

What is the source command (.) in Bash?

Answer:

source (or . followed by a space) executes a script in the current shell context, not a subshell. This means variables and functions defined in the sourced script become available in the current session. 'source ~/.bashrc' reloads the config.

Question 130

What is the difference between .bashrc and .bash_profile?

Answer:

.bash_profile (or .profile) is executed for login shells — when you log in via SSH or console. .bashrc is executed for interactive non-login shells — when you open a terminal emulator. Best practice: put environment settings in .bash_profile and source .bashrc from it.

Question 131

What is the PS1 variable in Bash?

Answer:

PS1 (Prompt String 1) defines the shell prompt appearance. 'PS1='\u@\h:\w\\$ "' shows 'username@hostname:path\$'. Escape codes: \u (username), \h (hostname), \w (working directory), \t (time), \\$ (# for root, \$ otherwise).

Question 132

What is process substitution?

Answer:

Process substitution allows a command's output to be used as a file. '<(command)' creates a named pipe. Example: 'diff <(sort file1) <(sort file2)' compares sorted versions of files without creating temporary files.

Question 133

What is the here document (here doc) in Bash?

Answer:

A heredoc allows multiline input to a command. 'cat << EOF ... EOF' sends everything between the markers to cat. Variables are expanded by default. Use '<<'EOF' (quoted) to prevent expansion. Useful for passing multiline strings to commands.

Question 134

What is job control in Bash?

Answer:

Job control manages multiple processes from one terminal. '&' runs a process in the background. 'jobs' lists background jobs. 'fg %1' brings job 1 to foreground. 'bg %1' resumes a stopped job in the background. Ctrl+Z suspends a running process.

Question 135

What is the nohup command?

Answer:

nohup (no hang up) runs a command immune to hangup signals, allowing it to continue after the user logs out. 'nohup command &' runs it in the background. Output is redirected to nohup.out unless otherwise specified.

Question 136

What is X Window System?

Answer:

The X Window System (X11 or X) is the foundational display system for graphical interfaces on Linux and Unix. It provides the network-transparent infrastructure for drawing windows, handling input devices, and rendering graphics. Display managers and desktop environments run on top of X.

Question 137

What is a display manager?

Answer:

A display manager is a graphical login screen that manages authentication and starts the desktop session. Examples include GDM (GNOME), SDDM (KDE), LightDM (used by many DEs), and XDM. It replaces the text-based console login for graphical systems.

Question 138

What is a desktop environment?

Answer:

A desktop environment provides a complete graphical interface including a window manager, taskbar, file manager, and bundled applications. Popular options for Linux include GNOME, KDE Plasma, XFCE, LXDE, Cinnamon, and MATE.

Question 139

What is the DISPLAY environment variable?

Answer:

DISPLAY specifies the X server to connect to for graphical output. Format: 'hostname:display_number.screen_number'. The default is ':0.0' or ':0', meaning the local display. Setting DISPLAY allows running GUI applications over SSH with X forwarding.

Question 140

What is Wayland in Linux?

Answer:

Wayland is a modern display server protocol designed to replace X11. It provides better security, performance, and simplicity by running compositing in the display server itself. GNOME and KDE have adopted Wayland as the default display server in recent versions.

Question 141

What is the difference between GNOME and KDE?

Answer:

GNOME is a desktop environment focused on simplicity, clean design, and ease of use, using GTK toolkit. KDE Plasma is feature-rich and highly customizable, using the Qt toolkit. GNOME suits users wanting simplicity; KDE suits users wanting extensive customization.

Question 142

What is a window manager?

Answer:

A window manager controls the placement, appearance, and management of windows on screen. Desktop environments include their own window manager (like Mutter for GNOME or KWin for KDE). Standalone lightweight window managers like i3, Openbox, and Fluxbox can also be used.

Question 143

How do you change the default desktop environment in Linux?

Answer:

Use the update-alternatives command to set the default session: 'update-alternatives --config x-session-manager'. Or choose the session type from the login screen (display manager). You can also modify the /etc/X11/default-display-manager file.

Question 144

What is xrandr?

Answer:

xrandr is a command-line tool for configuring screen resolution, rotation, and multi-monitor setups under X11. 'xrandr' lists displays and their modes. 'xrandr --output HDMI1 --mode 1920x1080' sets a specific resolution for a monitor.

Question 145

What is the ~/.xinitrc file?

Answer:

~/.xinitrc is a script executed when starting an X session using the startx command. It specifies which programs (like a window manager or desktop environment) to launch. Without a display manager, startx reads this file to start the graphical session.

Question 146

What is Accessibility in Linux desktops?

Answer:

Accessibility features help users with disabilities use their computers. Linux desktops provide tools like screen readers (Orca for GNOME), magnifiers, high-contrast themes, sticky keys, slow keys, and keyboard navigation. These are typically configured in system settings.

Question 147

What is a virtual console in Linux?

Answer:

Virtual consoles (or virtual terminals) are text-based login sessions that run independently. Linux typically has 6 virtual consoles (tty1-tty6) and one for the graphical session (tty7 on older systems). Switch with Ctrl+Alt+F1 through F6.

Question 148

What is the startx command?

Answer:

startx is a script that starts an X Window session from a virtual console. It reads ~/.xinitrc to determine which environment to start. 'startx' is typically used on minimal installations without a display manager to launch the graphical interface manually.

Question 149

What is the TERM variable?

Answer:

TERM specifies the type of terminal being used. It tells programs how to handle terminal capabilities like cursor movement, colors, and screen clearing. Common values: xterm-256color, vt100, screen. Programs like vim and top rely on TERM for correct rendering.

Question 150

What is the Locale in Linux?

Answer:

Locale defines the language, region, character encoding, and formatting rules for date, time, numbers, and currency. Common locale variables include LANG, LC_ALL, LC_MESSAGES. 'locale' shows current settings. 'localectl' manages locale configuration on systemd systems.

Section 5: Administrative Tasks

Question 151

How do you create a new user in Linux?

Answer:

Use 'useradd username' to create a user. Common options: -m (create home directory), -s (specify shell), -G (add to groups), -d (specify home directory). 'useradd -m -s /bin/bash -G sudo john' creates user john with home, bash shell, and sudo group access.

Question 152

How do you set or change a user's password?

Answer:

Use 'passwd username' to set/change a password. Running 'passwd' without arguments changes the current user's password. Root can change any user's password with 'passwd username'. 'passwd -l username' locks an account; '-u' unlocks it.

Question 153

How do you delete a user in Linux?

Answer:

'userdel username' deletes a user account. 'userdel -r username' also removes the home directory and mail spool. Use with caution as this permanently deletes user data. Check for running processes owned by the user before deletion.

Question 154

What is the /etc/passwd file?

Answer:

/etc/passwd stores user account information — one entry per user with 7 colon-separated fields: username:password:UID:GID:GECOS:home:shell. The password field typically shows 'x', indicating the actual password hash is in /etc/shadow.

Question 155

What is the /etc/shadow file?

Answer:

/etc/shadow stores encrypted password hashes and password aging information. It is readable only by root for security. Fields: username:hash:last_changed:min_age:max_age:warn:inactive:expire. If the hash field is '!', the account is locked.

Question 156

What is the `/etc/group` file?

Answer:

`/etc/group` defines groups and their members. Format: `group_name:password:GID:member_list`. Example: `'developers:x:1001:alice,bob'`. The 'x' means group passwords are in `/etc/gshadow`. Running `'groups username'` shows a user's group memberships.

Question 157

How do you add a user to a group?

Answer:

`'usermod -aG groupname username'` adds a user to a supplementary group without removing existing group memberships. '-a' means append. Example: `'usermod -aG docker john'` adds john to the docker group. The user must log out and back in for changes to take effect.

Question 158

What is the difference between primary and secondary groups?

Answer:

Every user has one primary group (defined in `/etc/passwd` as GID). New files are created with this group. Secondary (supplementary) groups give additional access. `'id username'` shows both primary and all secondary group memberships.

Question 159

What is the `chage` command?

Answer:

`chage` (change age) manages password aging policies. `'chage -l username'` lists current settings. `'chage -M 90 username'` sets max password age to 90 days. `'chage -d 0 username'` forces password change at next login. Used for security compliance.

Question 160

What is a cron job?

Answer:

A cron job is a scheduled task managed by the cron daemon (crond). Jobs are defined in crontab files with the format: `'minute hour day month weekday command'`. `'* * * * *'` runs every minute. `'0 2 * * 0'` runs at 2 AM every Sunday.

Question 161

How do you edit cron jobs?

Answer:

`'crontab -e'` opens the user's crontab for editing in the default editor. `'crontab -l'` lists current cron jobs. `'crontab -r'` removes all cron jobs. System-wide cron jobs are in `/etc/cron.d/`, `/etc/cron.daily/`, `/etc/cron.weekly/`, and `/etc/cron.monthly/`.

Question 162

What is at command?

Answer:

at schedules a one-time command to run at a specific time. 'at 3pm tomorrow' opens an interactive prompt where you type the command, then press Ctrl+D. 'atq' lists pending jobs. 'atrm job_number' removes a job. The atd daemon executes scheduled jobs.

Question 163

What is the anacron command?

Answer:

anacron runs jobs that were missed while the system was powered off. Unlike cron, anacron is designed for systems that don't run 24/7. Jobs are defined in /etc/anacrontab with periods in days. It ensures periodic maintenance jobs run after missed intervals.

Question 164

What is a system log and where are logs stored?

Answer:

System logs record system events, errors, and activities. In traditional systems, logs are in /var/log/: syslog or messages (general), auth.log (authentication), kern.log (kernel), dmesg (boot messages). On systemd systems, journalctl is used to view logs from journald.

Question 165

What is journalctl?

Answer:

journalctl queries and displays logs from the systemd journal. 'journalctl -u nginx' shows nginx service logs. 'journalctl -f' follows logs in real time. 'journalctl --since 'today'' shows today's logs. 'journalctl -p err' shows only error messages.

Question 166

What is syslog?

Answer:

Syslog is the standard logging protocol for Unix-like systems. The rsyslog or syslog-ng daemon collects log messages from the kernel and applications and writes them to files in /var/log/. Configuration is in /etc/rsyslog.conf or /etc/rsyslog.d/.

Question 167

What is the logrotate command?

Answer:

logrotate manages the automatic rotation, compression, and deletion of log files to prevent them from consuming too much disk space. Configuration is in /etc/logrotate.conf and /etc/logrotate.d/. It can be run automatically via cron or a systemd timer.

Question 168

What is the dmesg command?

Answer:

dmesg displays the kernel ring buffer messages, particularly useful for viewing hardware detection and driver messages at boot. 'dmesg | grep -i error' searches for errors. 'dmesg -T' shows human-readable timestamps. 'dmesg -c' clears the buffer.

Question 169

What is the systemctl command?

Answer:

systemctl controls the systemd init system. Common commands: 'systemctl start/stop/restart service', 'systemctl enable/disable service' (at boot), 'systemctl status service', 'systemctl list-units --type=service', 'systemctl daemon-reload' (after config changes).

Question 170

What is a systemd service unit file?

Answer:

A systemd unit file (.service) defines how to manage a service. Sections include [Unit] (description, dependencies), [Service] (ExecStart, ExecStop, Restart, User), and [Install] (WantedBy for enabling). Files are in /lib/systemd/system/ or /etc/systemd/system/.

Question 171

What is the difference between enable and start in systemctl?

Answer:

'systemctl start service' starts the service immediately but not at boot. 'systemctl enable service' configures it to start at boot but doesn't start it now. 'systemctl enable --now service' both enables and starts it immediately.

Question 172

What is the update-rc.d command?

Answer:

update-rc.d manages SysV init script links in /etc/rc?.d/ on Debian-based systems. 'update-rc.d service defaults' enables a service. 'update-rc.d service remove' disables it. On modern systems using systemd, systemctl is preferred.

Question 173

What is chkconfig?

Answer:

chkconfig is a utility used on Red Hat-based systems to manage SysV init scripts. 'chkconfig --list' shows services and their runlevel status. 'chkconfig httpd on' enables httpd for standard runlevels. Replaced by systemctl on modern systems.

Question 174

What is the `/etc/hosts` file?

Answer:

`/etc/hosts` maps hostnames to IP addresses locally, before DNS is queried. Format: 'IP hostname alias'. Example: '192.168.1.10 server1.example.com server1'. Adding entries here allows name resolution without a DNS server, useful for development and lab environments.

Question 175

What is the `su` command?

Answer:

'su' (switch user) allows switching to another user account. 'su - username' switches to username with a login shell (loads their environment). 'su -' switches to root. 'su - username -c 'command'' runs a single command as that user without interactive login.

Question 176

What is the `visudo` command?

Answer:

`visudo` safely edits the `/etc/sudoers` file, which controls `sudo` access. It validates the syntax before saving to prevent lock-out from `sudo`. Always use `visudo` instead of editing `/etc/sudoers` directly. Additional rules can be placed in `/etc/sudoers.d/`.

Question 177

What is the `/etc/sudoers` file?

Answer:

`/etc/sudoers` defines who can use `sudo` and what commands they can run. Format: 'user host=(runas) command'. 'alice ALL=(ALL) ALL' allows alice to run any command as any user. '%admin ALL=(ALL) ALL' applies to the admin group. Use `visudo` to edit.

Question 178

How do you check disk usage of a user's home directory?

Answer:

'du -sh /home/username' shows total disk usage for a user's home directory. 'du -ah /home/username | sort -rh | head -20' lists the 20 largest files/directories. 'quota -u username' shows quota usage if quotas are enabled.

Question 179

What is disk quota in Linux?

Answer:

Disk quotas limit the amount of disk space or number of files (inodes) a user or group can use. Managed with `quotacheck`, `quotaon`, `edquota`, and `repquota` commands. Quotas are configured in `/etc/fstab` with 'usrquota' and 'grpquota' mount options.

Question 180

What is the nice and renice command?

Answer:

nice sets the scheduling priority of a process when starting it. 'nice -n 10 command' runs with priority +10 (lower priority). renice changes priority of a running process: 'renice -n 5 -p PID'. Range is -20 (highest) to 19 (lowest). Only root can set negative values.

Question 181

What is the locale command?

Answer:

locale displays the current locale settings including language, character encoding, and regional formatting. 'locale -a' lists all available locales. 'localectl set-locale LANG=en_US.UTF-8' sets the system locale permanently on systemd systems.

Question 182

What is timezone configuration in Linux?

Answer:

Timezone is set in /etc/localtime (symlink to a zone file) and /etc/timezone (text file with zone name). Use 'timedatectl set-timezone America/New_York' on systemd systems. 'tzselect' is an interactive tool for selecting timezone. 'date' shows current time and timezone.

Question 183

What is the hwclock command?

Answer:

hwclock manages the hardware (BIOS/CMOS) clock. 'hwclock --show' displays the hardware clock time. 'hwclock --hctosys' sets system time from hardware clock. 'hwclock --systohc' sets hardware clock from system time. Use '--utc' to specify UTC.

Question 184

What is NTP (Network Time Protocol)?

Answer:

NTP synchronizes the system clock over the network with time servers. ntpd and chronyd are common NTP daemons. 'timedatectl set-ntp true' enables automatic time sync on systemd. 'chronyc tracking' shows synchronization status for chrony.

Question 185

What is the crontab syntax?

Answer:

Crontab has 5 time fields followed by the command: 'minute(0-59) hour(0-23) day(1-31) month(1-12) weekday(0-7) command'. Special strings: @reboot (at boot), @daily (once daily), @weekly (once weekly), @monthly (once monthly). '*' means every value.

Question 186

What is the purpose of `/etc/cron.allow` and `/etc/cron.deny`?

Answer:

`/etc/cron.allow` lists users permitted to use crontab. `/etc/cron.deny` lists users denied access. If `cron.allow` exists, only listed users can use cron. If only `cron.deny` exists, all users except those listed can use cron. If neither exists, only root can use cron.

Question 187

What is the systemd timer unit?

Answer:

systemd timers are an alternative to cron jobs. A `.timer` unit paired with a `.service` unit schedules tasks. `'systemctl list-timers'` shows all active timers. Timers support monotonic (relative) and realtime (calendar-based) scheduling with better logging than cron.

Question 188

How do you backup files in Linux?

Answer:

Common backup methods: `rsync` (`'rsync -avz source dest'`) for incremental backups, `tar` (`'tar -czf backup.tar.gz /data'`) for archives, `cp -r` for simple copies, and tools like Amanda, Bacula, or Duplicati for enterprise backups. Always test restores to verify backup integrity.

Question 189

What is `rsync`?

Answer:

`rsync` is a fast, versatile file copying and synchronization tool. It only transfers changed files, making it efficient for backups. `'rsync -avz /source/ user@remote:/dest/'` syncs to a remote host over SSH. `'--delete'` removes files at destination that no longer exist at source.

Question 190

What is the `cpio` command?

Answer:

`cpio` (copy in/out) copies files to and from archives. `'find . -print | cpio -o > archive.cpio'` creates an archive. `'cpio -i < archive.cpio'` extracts. `cpio` is used by the Linux kernel's `initramfs`. The `'-H newc'` format is used for `initrd` images.

Section 6: Essential System Services

Question 191

What is GRUB and what does it do?

Answer:

GRUB (GRand Unified Bootloader) is the default bootloader for most Linux distributions. It displays a menu at boot allowing you to select which OS or kernel to boot. Configuration is in `/etc/grub.d/` and `/etc/default/grub`. 'grub-mkconfig' generates the `grub.cfg` file.

Question 192

What is the boot process in Linux?

Answer:

The Linux boot process: 1) BIOS/UEFI initializes hardware, 2) Bootloader (GRUB) loads kernel, 3) Kernel initializes hardware and mounts `initramfs`, 4) Init system (`systemd`/`SysVinit`) starts, 5) Target/runlevel services start, 6) Login prompt appears.

Question 193

What is `initramfs`?

Answer:

`initramfs` (initial RAM filesystem) is a temporary root filesystem loaded into RAM by the bootloader. It contains drivers and tools needed to mount the real root filesystem, especially when it uses LVM, RAID, or encryption. After the real root is mounted, `initramfs` is discarded.

Question 194

What is UEFI and how does it differ from BIOS?

Answer:

UEFI (Unified Extensible Firmware Interface) is the modern replacement for BIOS. It supports GPT disks, secure boot, faster boot times, a graphical interface, and network booting. UEFI uses an EFI System Partition (ESP) while BIOS uses the MBR to store the bootloader.

Question 195

What is Secure Boot?

Answer:

Secure Boot is a UEFI feature that only allows software signed with trusted cryptographic keys to run during boot. It prevents malware from loading before the OS. Linux supports Secure Boot through signed bootloaders like `shim` and properly signed kernels.

Question 196

What is the kernel and how is it loaded?

Answer:

The Linux kernel is the core of the OS, managing hardware, memory, and processes. GRUB loads the kernel image (vmlinuz) into RAM along with initramfs. The kernel then decompresses itself, initializes devices, mounts initramfs as temporary root, and starts the init system.

Question 197

What is the `/boot/grub/grub.cfg` file?

Answer:

`grub.cfg` is the main GRUB configuration file containing boot menu entries, kernel parameters, and display settings. It is automatically generated by `'grub-mkconfig -o /boot/grub/grub.cfg'` or `'update-grub'`. You should not edit it directly – modify files in `/etc/default/grub` instead.

Question 198

How do you add kernel boot parameters in GRUB?

Answer:

Edit `/etc/default/grub` and add parameters to `GRUB_CMDLINE_LINUX` or `GRUB_CMDLINE_LINUX_DEFAULT`. Example: `'GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"'`. Then run `'update-grub'` or `'grub-mkconfig -o /boot/grub/grub.cfg'` to apply changes.

Question 199

What is single-user mode?

Answer:

Single-user mode (runlevel 1 or emergency target) boots the system with minimal services and gives a root shell without requiring a password on some systems. It is used for system recovery, password resets, and troubleshooting. Append `'single'` or `'1'` to kernel parameters in GRUB to enter it.

Question 200

What is PXE boot?

Answer:

PXE (Preboot Execution Environment) allows a computer to boot from a network server rather than local storage. It uses DHCP to get an IP address and TFTP to download a bootloader. Used for network installations (like installing Linux over the network) and diskless workstations.

Question 201

What is the SSH service?

Answer:

SSH (Secure Shell) provides encrypted remote access to Linux systems. The `sshd` daemon listens on port 22. Configuration is in `/etc/ssh/sshd_config`. Key features: password and key-based authentication, tunneling, and SFTP. `'systemctl status sshd'` checks its status.

Question 202

How do you configure SSH key-based authentication?

Answer:

1) Generate keys: 'ssh-keygen -t rsa -b 4096'. 2) Copy public key to server: 'ssh-copy-id user@server' or manually append ~/.ssh/id_rsa.pub to ~/.ssh/authorized_keys on the server. 3) Set permissions: 'chmod 700 ~/.ssh; chmod 600 ~/.ssh/authorized_keys'.

Question 203

What is the /etc/ssh/sshd_config file?

Answer:

sshd_config is the SSH server configuration file. Important settings: Port (default 22), PermitRootLogin (yes/no/prohibit-password), PasswordAuthentication (yes/no), PubkeyAuthentication (yes), AllowUsers/AllowGroups (restrict access). Restart sshd after changes.

Question 204

What is the scp command?

Answer:

scp (secure copy) transfers files between hosts over SSH. 'scp file user@host:/path/' copies a local file to remote. 'scp user@host:/path/file .' copies from remote to local. 'scp -r dir user@host:/path/' copies directories recursively. Speed is affected by encryption overhead.

Question 205

What is rsync over SSH?

Answer:

rsync can use SSH as the transport: 'rsync -avz -e ssh source user@host:/dest/'. This provides efficient incremental file transfer with SSH encryption. It is preferred over scp for syncing directories as it only transfers changed parts of files.

Question 206

What is a print system in Linux?

Answer:

CUPS (Common Unix Printing System) is the standard printing system for Linux and macOS. It provides a web interface (<http://localhost:631>), manages print queues, and supports many printer protocols. 'lp file.pdf' prints, 'lpq' shows queue, 'lprm' removes jobs.

Question 207

What is time synchronization and why is it important?

Answer:

Time synchronization ensures the system clock is accurate. This is critical for security (certificate validation, Kerberos), distributed systems, log correlation, and scheduled tasks. Linux uses NTP via ntpd or chronyd, or systemd-timesyncd for simple synchronization.

Question 208

What is chrony?

Answer:

chrony is a modern NTP implementation that synchronizes the system clock. It is particularly good for systems with intermittent network access. 'chronyc sources' shows time sources. 'chronyc tracking' shows synchronization status. Configuration is in `/etc/chrony.conf`.

Question 209

What is the ntpdate command?

Answer:

ntpdate immediately sets the system time from an NTP server. 'ntpdate pool.ntp.org' syncs the clock once. It is deprecated in favor of ntpd or chronyd for continuous synchronization. On modern systemd systems, 'timedatectl set-ntp true' is preferred.

Question 210

What is MTA (Mail Transfer Agent)?

Answer:

An MTA transfers email between systems using the SMTP protocol. Common Linux MTAs include Postfix, Sendmail, Exim, and qmail. 'mail' and 'sendmail' are command-line tools to send mail. MTAs deliver mail to local users (`/var/mail/`) or relay to other mail servers.

Question 211

What is Postfix?

Answer:

Postfix is a popular, secure, and efficient MTA for Linux. Its main config files are `/etc/postfix/main.cf` and `/etc/postfix/master.cf`. 'postfix start/stop/reload' manages the service. 'mailq' shows the mail queue. 'postsuper -d ALL' clears the queue.

Question 212

What is the mail command in Linux?

Answer:

mail is a command-line email client used to send and read email. 'echo 'Body' | mail -s 'Subject' user@example.com' sends an email. 'mail' without arguments opens the inbox. It uses the local MTA to send messages and reads from `/var/mail/username`.

Question 213

What is syslog facility and severity?

Answer:

Syslog messages have two properties: facility (source of the message: auth, kern, mail, daemon, user, local0-7) and severity (priority: emerg, alert, crit, err, warning, notice, info, debug). These are used in rsyslog configuration to route messages to appropriate log files.

Question 214

What is the `/etc/rsyslog.conf` file?

Answer:

`/etc/rsyslog.conf` configures the rsyslog daemon. It specifies rules for filtering and routing log messages. Format: 'facility.severity /path/to/logfile'. Example: 'auth.* /var/log/auth.log' logs all authentication messages. Additional configs go in `/etc/rsyslog.d/`.

Question 215

What is the logger command?

Answer:

logger sends messages to the system log (syslog). 'logger 'Backup completed'' adds a message to the syslog. Options: `-p` (priority, e.g., 'local0.info'), `-t` (tag/identifier). It is useful in scripts to log events to the system log for centralized monitoring.

Question 216

What is the purpose of `/etc/issue` and `/etc/motd`?

Answer:

`/etc/issue` displays a message before the login prompt (on virtual consoles and SSH). `/etc/motd` (Message of the Day) displays after successful login. Administrators use these to show system information, legal warnings, or maintenance notices to users.

Question 217

What is LDAP in Linux?

Answer:

LDAP (Lightweight Directory Access Protocol) is a protocol for accessing distributed directory services. It is used for centralized authentication and user management. Linux systems can authenticate against LDAP servers using PAM and NSS. OpenLDAP is the most common open-source LDAP implementation.

Question 218

What is PAM (Pluggable Authentication Modules)?

Answer:

PAM is a flexible authentication framework for Linux. It separates authentication from applications by using interchangeable modules. Configuration files are in `/etc/pam.d/`. PAM handles authentication (verifying identity), account (access control), session (setup), and password (management) operations.

Question 219

What is NSS (Name Service Switch)?

Answer:

NSS is a mechanism that controls where Linux looks for user/group information and hostname resolution. Configuration is in `/etc/nsswitch.conf`. Example: 'passwd: files ldap' means look in `/etc/passwd` first, then LDAP. It also controls DNS resolution order.

Question 220

What is the `/etc/nsswitch.conf` file?

Answer:

`/etc/nsswitch.conf` specifies the order of lookup for various services like user names, passwords, groups, and hostnames. Common databases: `passwd`, `group`, `shadow`, `hosts`, `networks`. Example: `'hosts: files dns'` checks `/etc/hosts` before querying DNS.

Question 221

What are the common GRUB2 commands for troubleshooting?

Answer:

In GRUB2 rescue mode: `'ls'` lists devices and partitions, `'set root=(hd0,1)'` sets the boot device, `'linux /vmlinuz root=/dev/sda1'` loads the kernel, `'initrd /initrd.img'` loads `initramfs`, `'boot'` boots the system. `'grub-install /dev/sda'` reinstalls GRUB to the MBR.

Question 222

What is the purpose of the `/etc/inittab` file?

Answer:

`/etc/inittab` was the main configuration file for `SysVinit`, defining the default runlevel and what actions to take at boot and during runlevel transitions. On modern `systemd`-based systems, this file is deprecated and not used; `systemd` targets replace runlevels.

Question 223

What is an email alias in Linux?

Answer:

Email aliases redirect mail sent to one address to another. System-wide aliases are in `/etc/aliases`. After editing, run `'newaliases'` to rebuild the alias database. Example: `'root: admin@example.com'` forwards root's mail to the admin. User-level aliases are in `~/.forward`.

Section 7: Networking Fundamentals

Question 224

What is an IP address?

Answer:

An IP address is a numerical label assigned to each device on a network. IPv4 addresses are 32-bit (e.g., 192.168.1.1). IPv6 addresses are 128-bit (e.g., 2001:db8::1). IP addresses identify hosts and enable routing of packets across networks.

Question 225

What is a subnet mask?

Answer:

A subnet mask determines which part of an IP address is the network portion and which is the host portion. Example: 255.255.255.0 (or /24) means the first 24 bits are the network. It is used by routers to determine if a destination is local or remote.

Question 226

What is CIDR notation?

Answer:

CIDR (Classless Inter-Domain Routing) notation expresses an IP address and subnet mask together. '/24' means 24 bits are network bits, equivalent to subnet 255.255.255.0. '192.168.1.0/24' represents the entire 192.168.1.x subnet with 254 usable hosts.

Question 227

What is a default gateway?

Answer:

The default gateway is the router that a device sends traffic to when the destination is not on the local network. It routes packets to other networks. Set with 'ip route add default via GATEWAY_IP'. Typically the first or last host address in a subnet.

Question 228

What is DNS?

Answer:

DNS (Domain Name System) translates human-readable domain names (like google.com) into IP addresses. It works hierarchically: Root servers → TLD servers (.com, .org) → Authoritative servers. DNS uses UDP port 53 primarily. Linux uses /etc/resolv.conf to configure DNS servers.

Question 229

What is the `/etc/resolv.conf` file?

Answer:

`/etc/resolv.conf` configures DNS resolution. It specifies nameservers ('nameserver 8.8.8.8'), search domains ('search example.com'), and lookup options. On modern systems with NetworkManager or systemd-resolved, this file is managed automatically.

Question 230

What is the `ip` command?

Answer:

`ip` is the modern replacement for `ifconfig`, `route`, and `arp`. '`ip addr show`' displays IP addresses. '`ip link set eth0 up/down`' enables/disables an interface. '`ip route show`' displays the routing table. '`ip addr add 192.168.1.10/24 dev eth0`' assigns an IP.

Question 231

What is the `ifconfig` command?

Answer:

`ifconfig` (interface configuration) is a legacy command for displaying and configuring network interfaces. '`ifconfig`' shows active interfaces. '`ifconfig eth0 192.168.1.10 netmask 255.255.255.0`' assigns an IP. Replaced by the '`ip`' command but still widely used.

Question 232

What is the `route` command?

Answer:

`route` displays or modifies the IP routing table. '`route -n`' shows the routing table with numeric addresses. '`route add default gw 192.168.1.1`' adds a default gateway. Replaced by '`ip route`' but still commonly found on older systems.

Question 233

What is the `netstat` command?

Answer:

`netstat` displays network connections, routing tables, interface statistics, and socket information. '`netstat -tuln`' shows listening TCP/UDP ports. '`netstat -an`' shows all connections. Replaced by '`ss`' on modern systems but still widely used.

Question 234

What is the `ss` command?

Answer:

`ss` (socket statistics) is the modern replacement for `netstat`. It is faster and provides more information. '`ss -tuln`' shows listening TCP/UDP ports. '`ss -tp`' shows TCP connections with process names. '`ss -s`' shows socket summary statistics.

Question 235

What is the ping command?

Answer:

ping sends ICMP echo requests to test connectivity to a host. 'ping hostname' sends continuous pings (Ctrl+C to stop). 'ping -c 4 host' sends 4 packets. Output shows round-trip time and packet loss. Blocked by firewalls does not necessarily mean host is down.

Question 236

What is traceroute?

Answer:

traceroute (or tracepath) shows the path packets take to reach a destination, displaying each hop (router) along the way with latency. 'traceroute google.com' maps the route. On Windows it's tracert. Useful for diagnosing where network delays or failures occur.

Question 237

What is the nslookup command?

Answer:

nslookup queries DNS servers for information about domain names or IP addresses. 'nslookup google.com' looks up the IP for google.com. 'nslookup 8.8.8.8' does a reverse lookup. The interactive mode allows multiple queries with different servers.

Question 238

What is the dig command?

Answer:

dig (Domain Information Groper) is a flexible DNS query tool. 'dig google.com' queries A records. 'dig google.com MX' queries mail exchange records. 'dig @8.8.8.8 google.com' queries a specific server. 'dig +short google.com' shows just the answer.

Question 239

What is DHCP?

Answer:

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, gateways, and DNS servers to clients. The DHCP server maintains a pool of addresses. 'dhclient eth0' requests a DHCP address. DHCPd or isc-dhcp-server is a common Linux DHCP server.

Question 240

What is a network interface in Linux?

Answer:

A network interface is the connection point between the system and the network. Physical interfaces: eth0, enp2s0 (Ethernet), wlan0, wlp3s0 (WiFi). Virtual: lo (loopback, 127.0.0.1), docker0, virbr0, tun0 (VPN). 'ip link show' lists all interfaces.

Question 241

What is the loopback interface?

Answer:

The loopback interface (lo) is a virtual network interface with IP 127.0.0.1 (localhost). Traffic sent to it is looped back to the same host without leaving the system. It is used for inter-process communication and testing network applications locally.

Question 242

What are TCP and UDP?

Answer:

TCP (Transmission Control Protocol) is connection-oriented, ensuring reliable, ordered delivery with error checking. UDP (User Datagram Protocol) is connectionless, faster but unreliable — no guarantee of delivery or order. TCP is used for HTTP, SSH, FTP; UDP for DNS, DHCP, streaming.

Question 243

What is a port number?

Answer:

A port number identifies a specific application or service on a host. Range: 0–65535. Well-known ports (0–1023): reserved for standard services (HTTP=80, HTTPS=443, SSH=22, FTP=21, DNS=53). Registered ports (1024–49151): used by applications. Dynamic ports (49152–65535): temporary.

Question 244

What is the /etc/services file?

Answer:

/etc/services maps port numbers and protocols to service names. It is used by many network applications to look up port numbers by name. Example: 'ssh 22/tcp' means SSH uses TCP port 22. This file does not control which services run — it's just a reference.

Question 245

What is a firewall?

Answer:

A firewall filters network traffic based on rules, blocking or allowing connections. Linux uses iptables or nftables for packet filtering. firewalld (on RHEL/CentOS) and ufw (on Ubuntu) provide higher-level interfaces. Rules can filter by IP, port, protocol, and direction.

Question 246

What is iptables?

Answer:

iptables is the traditional Linux packet filtering framework. It uses tables (filter, nat, mangle) with chains (INPUT, OUTPUT, FORWARD). 'iptables -A INPUT -p tcp --dport 22 -j ACCEPT' allows SSH. 'iptables -L -n -v' lists current rules. Rules are not persistent across reboots without saving.

Question 247

What is nftables?

Answer:

nftables is the modern replacement for iptables, offering better performance and a cleaner syntax. It uses a single framework instead of separate iptables, ip6tables, arptables commands. 'nft list ruleset' shows rules. 'nft add rule ip filter input tcp dport 22 accept' allows SSH.

Question 248

What is NetworkManager?

Answer:

NetworkManager is a daemon that manages network connections on Linux. It handles wired, wireless, VPN, and mobile connections. 'nmcli' is the command-line interface. 'nmtui' provides a text-based UI. It is the default network management tool on most modern desktop and server distributions.

Question 249

What is the nmcli command?

Answer:

nmcli (NetworkManager CLI) manages network connections. 'nmcli device status' shows devices. 'nmcli connection show' lists connections. 'nmcli connection add type ethernet ...' creates a connection. 'nmcli connection up/down name' activates/deactivates. Used to configure static IPs, VPNs, and WiFi.

Question 250

What is the /etc/network/interfaces file?

Answer:

/etc/network/interfaces is the network configuration file on Debian/Ubuntu systems not using NetworkManager. It defines static IP, DHCP, and interface settings. 'ifup eth0' brings up an interface; 'ifdown eth0' brings it down using this config file.

Question 251

What is hostname resolution order?

Answer:

Linux resolves hostnames in the order defined in /etc/nsswitch.conf under the 'hosts' entry. Typical: 'hosts: files dns myhostname' — checks /etc/hosts first, then DNS servers in /etc/resolv.conf. You can add MDNS, WINS, or other methods as additional lookups.

Question 252

What is the hostname command?

Answer:

'hostname' displays or sets the system's hostname. 'hostname' shows the current hostname. 'hostname newname' temporarily changes it (not persistent). For permanent changes: 'hostnamectl set-hostname newname' on systemd, or edit /etc/hostname and /etc/hosts.

Question 253

What is TCP/IP model?

Answer:

The TCP/IP model has 4 layers: Application (HTTP, FTP, DNS — corresponds to OSI layers 5–7), Transport (TCP, UDP — OSI layer 4), Internet (IP, ICMP — OSI layer 3), and Network Access/Link (Ethernet, WiFi — OSI layers 1–2). It describes how data is packaged and transmitted across networks.

Question 254

What is ICMP?

Answer:

ICMP (Internet Control Message Protocol) is used for error reporting and diagnostics in IP networks. ping uses ICMP echo requests. traceroute uses ICMP time-exceeded messages. It is not used for data transfer — it is a control/diagnostic protocol operating at the network layer.

Question 255

What is ARP?

Answer:

ARP (Address Resolution Protocol) maps IP addresses to MAC addresses within a local network. When a host wants to communicate with an IP, it broadcasts an ARP request asking 'Who has this IP?'. The owner replies with its MAC address. 'arp -n' shows the ARP cache.

Question 256

What is NAT (Network Address Translation)?

Answer:

NAT allows multiple devices on a private network to share a single public IP address. The router translates private IP addresses to the public IP for outgoing traffic and reverses the translation for incoming responses. iptables with MASQUERADE is used for NAT on Linux.

Question 257

What is a VPN?

Answer:

A VPN (Virtual Private Network) creates an encrypted tunnel over a public network, allowing secure remote access. Linux supports OpenVPN, WireGuard, IPSec/L2TP, and PPTP. VPNs create virtual network interfaces (tun0, wg0) that route traffic through the encrypted tunnel.

Question 258

What is network bonding in Linux?

Answer:

Network bonding combines multiple physical network interfaces into a single logical interface for redundancy or increased bandwidth. The bonding kernel module creates a bond0 interface. Modes include active-backup (failover), round-robin (load balancing), and LACP (802.3ad). Config: /etc/network/interfaces or NetworkManager.

Question 259

What is the curl command?

Answer:

curl transfers data from or to a server using various protocols (HTTP, HTTPS, FTP, etc.). 'curl https://example.com' downloads a webpage. 'curl -o file.zip URL' saves to a file. 'curl -X POST -d 'data' URL' sends POST data. '-H' adds headers. Widely used for API testing.

Question 260

What is the wget command?

Answer:

wget downloads files from the web (HTTP, HTTPS, FTP). 'wget URL' downloads to the current directory. 'wget -O filename URL' saves with a specific name. 'wget -r URL' downloads recursively. 'wget -c URL' resumes an interrupted download. Useful for scripted downloads.

Question 261

What is the netcat (nc) command?

Answer:

netcat (nc) is a versatile networking utility for reading and writing data across TCP/UDP connections. 'nc -l 8080' listens on port 8080. 'nc host 80' connects to port 80. Used for port scanning, file transfer, and as a simple server/client. Often called the 'TCP/IP Swiss Army knife'.

Question 262

What is the tcpdump command?

Answer:

tcpdump is a command-line packet analyzer that captures and analyzes network traffic. 'tcpdump -i eth0' captures on eth0. 'tcpdump -i eth0 port 80' filters HTTP. 'tcpdump -w capture.pcap' saves to file for analysis in Wireshark. Requires root or CAP_NET_ADMIN capability.

Question 263

What is mtr?

Answer:

mtr (Matt's Traceroute) combines ping and traceroute into a single diagnostic tool. It continuously sends packets to each hop and displays packet loss and latency statistics in real time. 'mtr google.com' shows a live updating table of each hop on the route to google.com.

Section 8: Security

Question 264

What is SELinux?

Answer:

SELinux (Security-Enhanced Linux) is a mandatory access control (MAC) security system built into the Linux kernel. It enforces policies that restrict what processes can do, limiting damage from security breaches. Modes: Enforcing, Permissive (log only), Disabled. Common on RHEL/CentOS.

Question 265

What is AppArmor?

Answer:

AppArmor is a Linux security module providing MAC by confining programs to a set of allowed operations using profiles. Unlike SELinux (label-based), AppArmor uses path-based policies. It is the default on Ubuntu and SUSE. 'aa-status' shows status; profiles are in /etc/apparmor.d/.

Question 266

What is file encryption in Linux?

Answer:

Linux supports several encryption tools: LUKS/cryptsetup for full disk encryption, GPG for file/email encryption, and SSL/TLS for network encryption. 'gpg -c file' encrypts a file symmetrically. 'gpg --gen-key' creates a key pair. LUKS encrypts entire block devices.

Question 267

What is GPG?

Answer:

GPG (GNU Privacy Guard) is an implementation of the OpenPGP standard for encryption and signing. 'gpg --gen-key' creates a key pair. 'gpg -e -r recipient file' encrypts for a recipient. 'gpg -d file.gpg' decrypts. 'gpg --sign file' creates a digital signature.

Question 268

What is LUKS?

Answer:

LUKS (Linux Unified Key Setup) is the standard for Linux disk encryption. It encrypts entire partitions or disks. 'cryptsetup luksFormat /dev/sdb1' sets up encryption. 'cryptsetup open /dev/sdb1 mydata' opens the device. Mount the mapped device at /dev/mapper/mydata.

Question 269

What is SSH port forwarding?

Answer:

SSH port forwarding tunnels network traffic through an SSH connection. Local forwarding: `'ssh -L local_port:remote_host:remote_port user@ssh_server'` forwards local traffic to a remote destination. Remote forwarding: `'ssh -R'` exposes a local port on the remote server.

Question 270

What is fail2ban?

Answer:

fail2ban monitors log files for repeated failed authentication attempts and automatically bans offending IP addresses using iptables or firewalld rules. Configuration in `/etc/fail2ban/jail.conf`. `'fail2ban-client status sshd'` shows banned IPs. Used to protect SSH, Apache, and other services.

Question 271

What is TCP Wrappers?

Answer:

TCP Wrappers provide host-based access control for network services. `/etc/hosts.allow` defines allowed hosts; `/etc/hosts.deny` defines denied hosts. Rules are evaluated in order: allow first, then deny. Example: `'sshd: 192.168.1.0/24'` in `hosts.allow` allows SSH from the subnet.

Question 272

What is the umask in Linux?

Answer:

umask (user file creation mask) sets default permissions for new files and directories by masking out bits from the maximum (666 for files, 777 for directories). `'umask 022'` results in files with 644 (rw-r--r--) and directories with 755 (rwxr-xr-x). Set in `~/.bashrc` for users.

Question 273

What is SUID (Set User ID)?

Answer:

SUID is a special permission bit that, when set on an executable, runs the program with the permissions of the file owner rather than the user running it. `'/bin/passwd'` is a common SUID example — it runs as root to modify `/etc/shadow`. Set with `'chmod u+s file'` or mode 4xxx.

Question 274

What is SGID (Set Group ID)?

Answer:

SGID on an executable runs it with the file's group permissions instead of the user's group. When set on a directory, new files created inside inherit the directory's group. Set with `'chmod g+s'`. Used for collaborative directories where files should all have the same group.

Question 275

What is the sticky bit?

Answer:

The sticky bit on a directory prevents users from deleting or renaming files they don't own, even if they have write permission to the directory. `/tmp` uses this. Set with `'chmod +t dir'` or mode `1xxx`. In `'ls -l'`, it shows as `'t'` in the others execute position (e.g., `drwxrwxrwt`).

Question 276

What are ACLs (Access Control Lists)?

Answer:

ACLs provide fine-grained permission control beyond the standard owner/group/other model. `'setfacl -m u:alice:rw file'` grants alice read/write access. `'getfacl file'` shows ACL entries. ACLs require filesystem support (ext4, xfs support them) and the `'acl'` mount option.

Question 277

What is the openssl command?

Answer:

openssl is a toolkit implementing SSL/TLS and general cryptographic functions. `'openssl genrsa -out key.pem 2048'` generates an RSA key. `'openssl req -new -x509 -key key.pem -out cert.pem'` creates a self-signed certificate. `'openssl s_client -connect host:443'` tests SSL connections.

Question 278

What is a digital certificate?

Answer:

A digital certificate is an electronic document that binds a public key to an identity (domain, organization) and is signed by a Certificate Authority (CA). Used in HTTPS, email, and VPNs. `'openssl x509 -in cert.pem -text'` displays certificate details.

Question 279

What is the difference between symmetric and asymmetric encryption?

Answer:

Symmetric encryption uses the same key for encryption and decryption (AES, DES). It is fast but requires secure key exchange. Asymmetric encryption uses a key pair — public key to encrypt, private key to decrypt (RSA, ECC). It is slower but solves key distribution. HTTPS uses both.

Question 280

What is sudo configuration best practices?

Answer:

Best practices: Grant minimal privileges (NOPASSWD only when necessary), use `'visudo'` always, create sudo rules for specific commands rather than ALL, use groups (`%groupname`) for management, enable logging (Defaults logfile), restrict to necessary hosts, and audit sudo usage regularly.

Question 281

What is the last command?

Answer:

'last' shows a list of recent logins and reboots from /var/log/wtmp. 'lastb' shows failed login attempts from /var/log/btmp. 'last -n 20' shows the last 20 entries. Useful for security auditing to identify who has logged in and when.

Question 282

What is the who and w command?

Answer:

'who' shows currently logged-in users with terminal and login time. 'w' provides more detail including idle time, CPU usage, and current process for each logged-in user. 'whoami' shows the current user's name. These are useful for monitoring active sessions.

Question 283

What is /etc/security/limits.conf?

Answer:

/etc/security/limits.conf sets resource limits for users and groups managed by PAM. Limits include: nofile (open files), nproc (max processes), memlock (locked memory), core (core dump size). 'ulimit -a' shows current limits. Hard limits are set by root; users can lower their soft limits.

Question 284

What is port scanning?

Answer:

Port scanning probes network hosts to discover open ports and services. nmap is the most popular tool: 'nmap -sV host' scans and identifies service versions. 'nmap -sn 192.168.1.0/24' pings a subnet. Understanding port scanning helps in security hardening and identifying exposed services.

Question 285

What is nmap?

Answer:

nmap (Network Mapper) is a powerful network scanning and security auditing tool. Common uses: 'nmap -sS host' (SYN scan), 'nmap -O host' (OS detection), 'nmap -sV host' (service version), 'nmap -p 1-1000 host' (specific port range), 'nmap -A host' (aggressive, all detection features).

Question 286

What is the difference between authentication and authorization?

Answer:

Authentication verifies identity — confirming who you are (username/password, SSH key, biometrics). Authorization determines what an authenticated user is allowed to do (file permissions, sudo rules, ACLs). Authentication always comes first; authorization is the access control that follows.

Question 287

What are common security hardening practices?

Answer:

Key practices: disable unused services, use SSH key authentication and disable root login, keep software updated (unattended-upgrades), configure firewall rules, enable SELinux/AppArmor, monitor logs (fail2ban), use strong passwords and MFA, encrypt sensitive data, and regularly audit user accounts and permissions.

Question 288

What is the chroot command?

Answer:

chroot changes the apparent root directory for a process and its children, isolating it from the rest of the filesystem. 'chroot /newroot /bin/bash' opens a shell confined to /newroot. Used for system recovery (boot from live media, chroot to broken system) and basic sandboxing.

Question 289

What is the find command for security auditing?

Answer:

find is useful for security: 'find / -perm -4000' finds SUID files, 'find / -perm -2000' finds SGID files, 'find / -nouser' finds files with no valid owner (orphaned), 'find / -perm -o+w' finds world-writable files. These help identify potential security vulnerabilities.

Question 290

What is password hashing?

Answer:

Password hashing converts a plain text password into a fixed-length hash using algorithms like SHA-512, bcrypt, or Argon2. Linux stores password hashes in /etc/shadow. The hash includes a salt (random value) to prevent rainbow table attacks. 'passwd' automatically handles hashing.

Question 291

What is two-factor authentication (2FA) in Linux?

Answer:

2FA requires two forms of verification: something you know (password) and something you have (TOTP app like Google Authenticator) or something you are (biometrics). Linux SSH can use 2FA with the 'libpam-google-authenticator' PAM module configured in /etc/pam.d/sshd.

Question 292

What is the auditd daemon?

Answer:

auditd (Linux Audit Daemon) records security-relevant system events to /var/log/audit/audit.log. Configured via audit rules ('auditctl -w /etc/passwd -p wa' watches the passwd file). 'ausearch' searches audit logs. 'aureport' generates reports. Used for compliance (PCI-DSS, HIPAA) and forensics.

Question 293

What is intrusion detection in Linux?

Answer:

Intrusion Detection Systems (IDS) monitor systems for suspicious activity. Host-based IDS (HIDS) tools for Linux include AIDE (file integrity monitoring — 'aide --check'), Tripwire, and OSSEC. Network-based IDS (NIDS) tools include Snort and Suricata that analyze network traffic for attack signatures.

Question 294

What are best practices for SSH security?

Answer:

SSH security best practices: disable root login (PermitRootLogin no), use key-based auth only (PasswordAuthentication no), change default port (Port 2222), restrict users (AllowUsers), use fail2ban, enable 2FA, use strong key algorithms (Ed25519), limit login attempts (MaxAuthTries 3), and keep OpenSSH updated.
